

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary End Semester Examination – Winter 2022-23 Course: B. Tech. Branch: Mechanical Engg. Semester : III Subject Code & Name: Thermodynamics (BTMEC305) Max Marks: 60 Date: __/01/2023 Duration: 3 Hr.															
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam table is permitted.															
Q. 1	Attempt the following	(Level/CO)	Marks												
A)	Describe various types of thermodynamic equilibrium.	Understand	4												
B)	In a steam power plant, steam flows steadily from the steam generator to the turbine. In order to increase the velocity of stream it passes through a horizontal nozzle having an inlet area of 0.1 m^2. Steam at 1000 kPa and 400°C enters the nozzle with a velocity of 70 m/s. The steam leaves the nozzle at 300 kPa and 300°C and there is a heat loss of 30 kW while steam passing through it. Properties of steam can be referred from the table: <table border="1"> <thead> <tr> <th>Pressure (kPa)</th><th>Temperature ($^\circ\text{C}$)</th><th>Enthalpy (kJ/kg)</th><th>Specific Volume (m^3/kg)</th></tr> </thead> <tbody> <tr> <td>1000</td><td>400</td><td>3264</td><td>0.30</td></tr> <tr> <td>300</td><td>300</td><td>3069.3</td><td>0.88</td></tr> </tbody> </table> Determine the mass flow rate of steam, the exit velocity and the exit area of the nozzle.	Pressure (kPa)	Temperature ($^\circ\text{C}$)	Enthalpy (kJ/kg)	Specific Volume (m^3/kg)	1000	400	3264	0.30	300	300	3069.3	0.88	Evaluate	8
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Q.2	Attempt any two questions A/B/C														
A)	Describe the three laws of thermodynamics: 1) Zeroth Law, 2) First law and 3) Statements of second law of thermodynamics.	Understand	6												
B)	Describe the following terms: 1) Types of system, 2) Control Volume, 3) Homogeneous and heterogeneous System, 4) Extensive and Intensive properties, along with practical examples.	Understand	6												
C)	A rigid tank contains 1 m^3 of air at 200 kPa and at 50°C . How much heat must be transferred to the air to raise the air pressure to 400 kPa ? Air can be considered as an ideal gas and the value of constant volume specific heat (C_v) for air at 210°C is $0.742 \text{ kJ/kg } ^\circ\text{C}$.	Analyze and Evaluate	6												
Q. 3	Attempt any two of the following														
A)	A cyclic heat engine operates between a source temperature of 800°C and a sink temperature of 30°C . What is the least rate of heat rejection for 1 kW as net work output of the engine?	Evaluate	6												
B)	Describe the Inequality of Clausius and the three criteria/conditions of the reversibility of a cycle.	Remember /Understand	6												
C)	Describe the applications of entropy principle. Also describe the phenomenon of entropy transfer with the heat flow.	Evaluate	6												
Q.4	Attempt any two of the following														
A)	Describe the terminology: i) High and Low grade energy, ii) Available and Unavailable energy, iii) Dead state	Understand	6												

B)	What is Quality or Dryness fraction of steam. Describe any one calorimeter in detail which can measure the quality of steam, with the help of neat schematic and mathematical expressions.	Understand	6
C)	Steam initially at 0.3 MPa, 250°C is cooled at constant volume. (i) at what temperature will the steam become saturated vapour ? (ii) What is the quality at 80°C ? What is the heat transferred per kg of steam in cooling from 250°C to 80°C.	Remember	6
Q. 5 Attempt the following			
A)	A fluid at 200 kPa and 300°C has a volume of 0.8 m ³ . In a frictionless process at constant volume the pressure changes to 100 kPa. Find the final temperature and the heat transferred if the fluid is air. Also find the properties if the fluid is steam.	Analyze and Evaluate	6
B)	State and describe the various process steam can undergo in a cycle with the help of P-h (Pressure –Enthalpy) and or T-S (Temperature-Entropy) Charts.	Understand	6
*** End ***			

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